Perceptual Organization (II)

Introduction to Computational and Biological Vision

CS 202-1-5261

Computer Science Department, BGU

Ohad Ben-Shahar

Perceptual Organization

Why do things *look* they way they do? [Koffka 1935]

External (Environment) vs. Internal (organism) influences

- **Because the world is the way it is.**
- **Because we are (our nervous systems) are they way they are.**

Empiricism vs. Nativism

- **Because we learned to see them this way.**
- **Because we are programmed to see them this way (learned through evolution).**

Atomism vs. Holism

- **Because of the way each small piece of the visual field appears (local properties and influence).**
- **Because of the way the whole visual field is organized (global influence over local properties).**

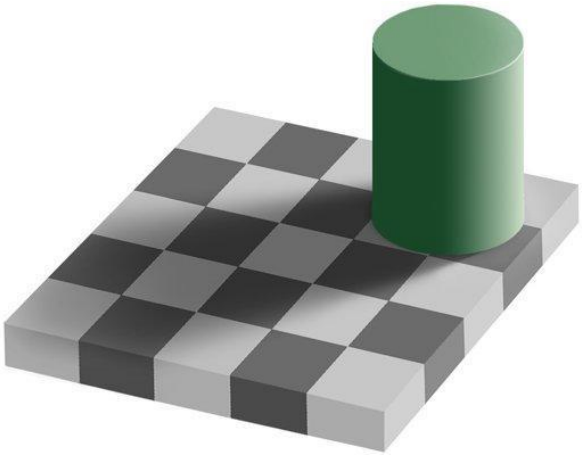
Perceptual Organization

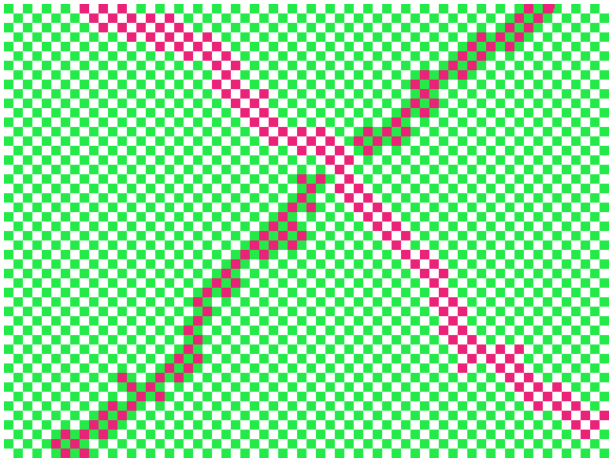
Classical theories of visual perception

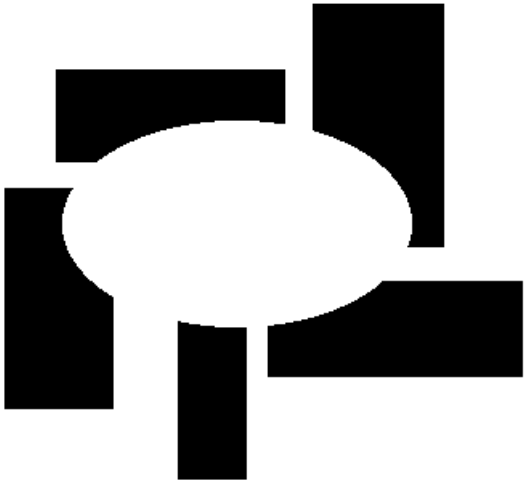
Theory	Environment vs. Organism	Empiricism Vs. Nativism	Atomism Vs. Holism
Structuralism	Organism	Empiricism	Atomism
Gestaltism	Organism	Nativism	Holism
Ecological Optics	Environment	Nativism	Holism

Perceptual Organization

Why do things *look* they way they do? [Koffka 1935]

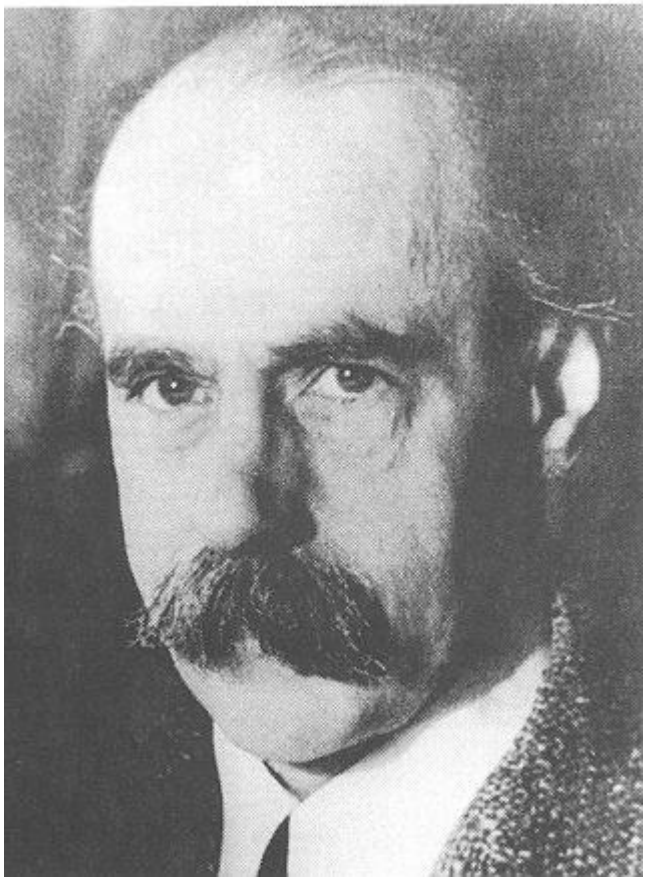






Perceptual Organization

What is Perceptual Organization



I stand at the window and see a house, trees, sky.

Theoretically I might say there were 327 brightnesses and nuances of colour. Do I have "327"? No. I have sky, house, and trees. *It is impossible to achieve "327" as such*.

Max Wertheimer, 1923

Perceptual Organization

What is Perceptual Organization

…[The process by which the] bits and pieces of visual information are structured into the larger units of perceived objects and their interrelations.

Stephen Palmer, 1999

…Perceptual organization is the process by which particular relationship (selected from alternative relationships) among potentially separate elements are perceived and guide the interpretation of those elements...

Pomerantz & Kubovy, 1986

Grouping

Why Perceptual Organization?

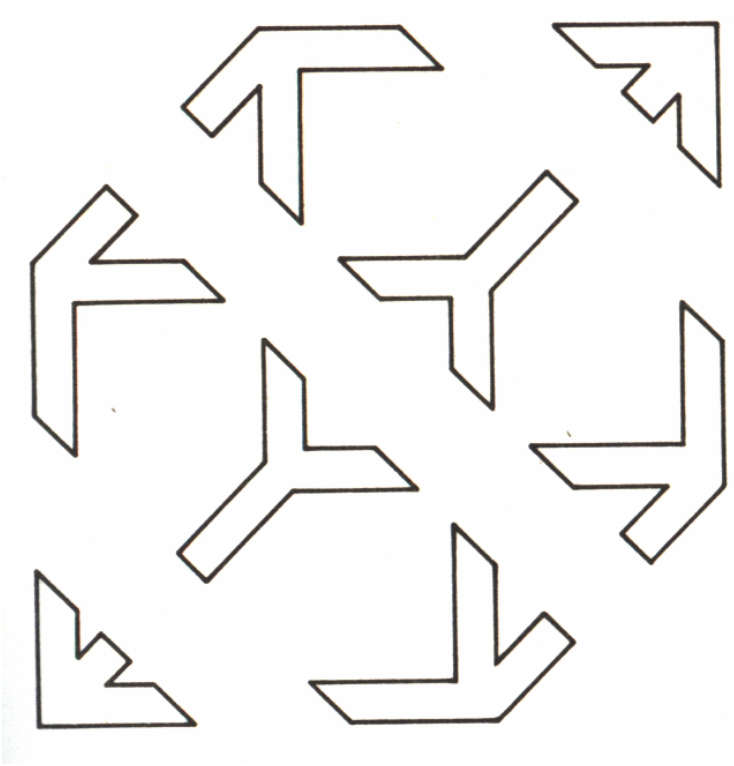
People's ability to perceive structure in images exists apart from both the perception of tri-dimensionality and from the recognition of familiar objects. That is, we organize the data even when we have no idea what it is we are organizing. What is remarkable is the degree to which such naively perceived structure survives more or less intact once a semantic context is established ... It is almost as if the visual system has some basis for guessing what is important without knowing why.

We argue that the aim of perceptual organization is the discovery and description of spatio-temporal coherence and regularity... A description that decomposes the image into constituents that capture regularity or coherence therefore provides descriptive chunks that act as "semantic precursors", in the sense that they deserve or demand explanation. .. By anticipating semantically important relationships, it provides a basis for interpretation in terms of stored models, and for learning new model.

Witkin & Tenenbaum, 1983

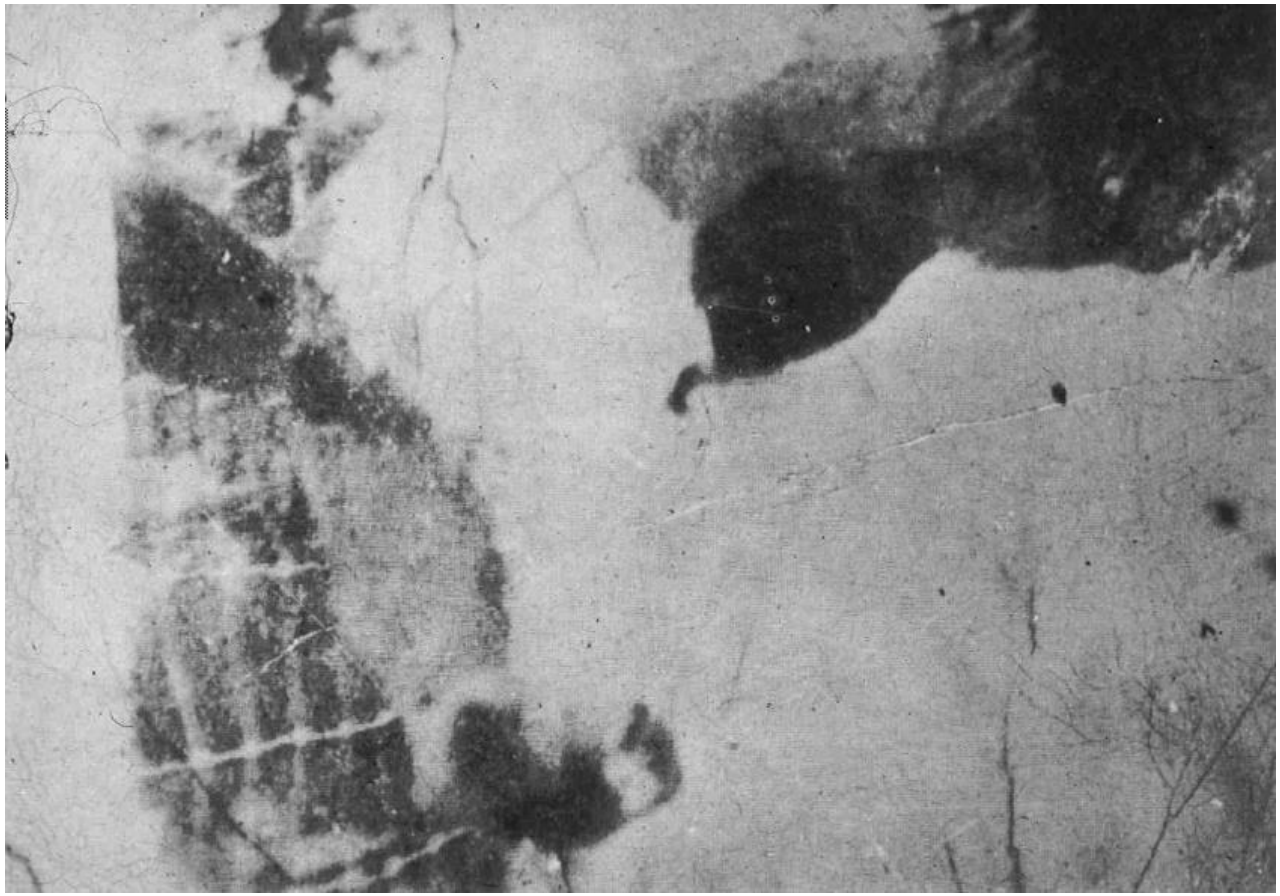
Perceptual Organization

Why Perceptual Organization?



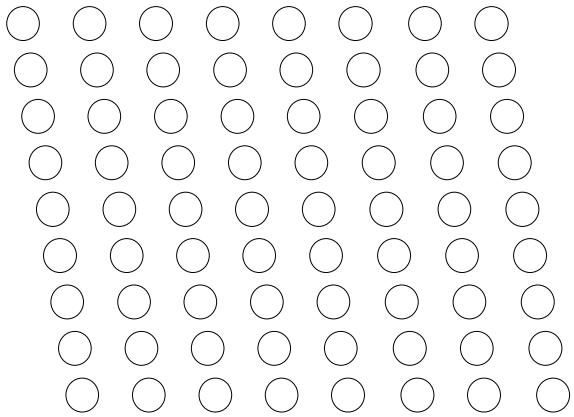
Perceptual Organization

Why Perceptual Organization?

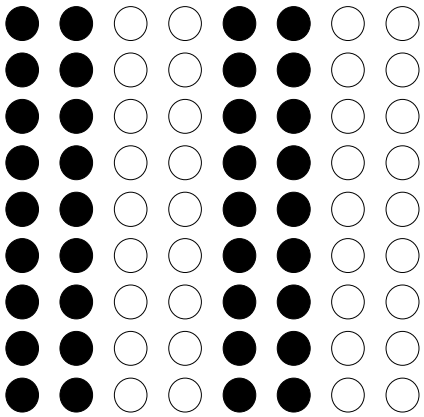


Perceptual Organization

Gestalt Laws of Perceptual Organization



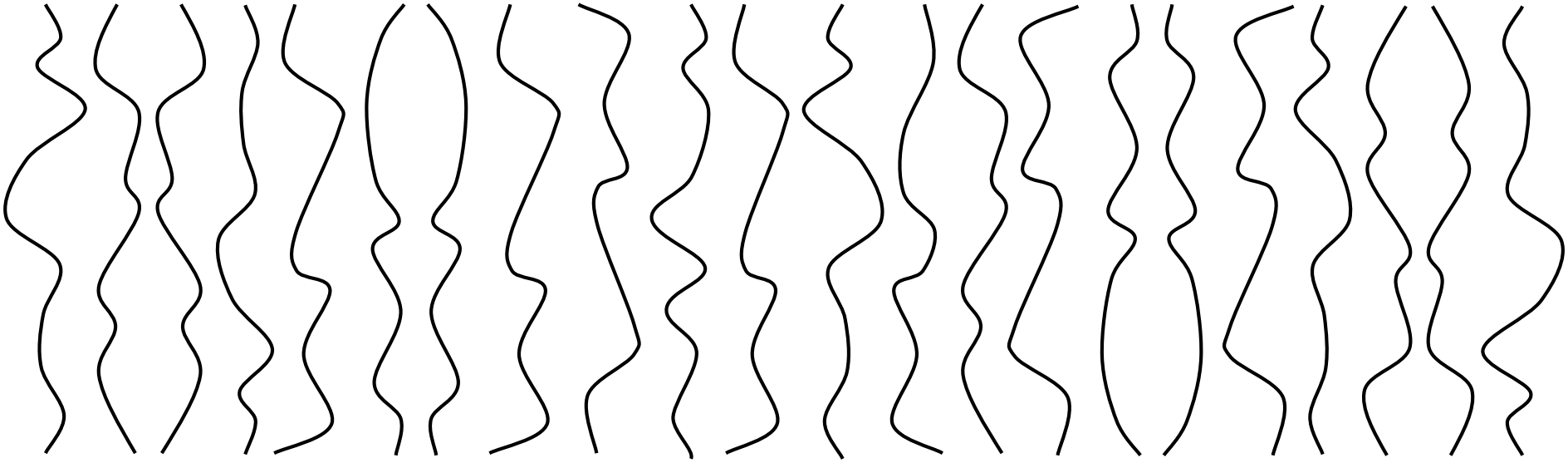
Proximity



Similarity

Perceptual Organization

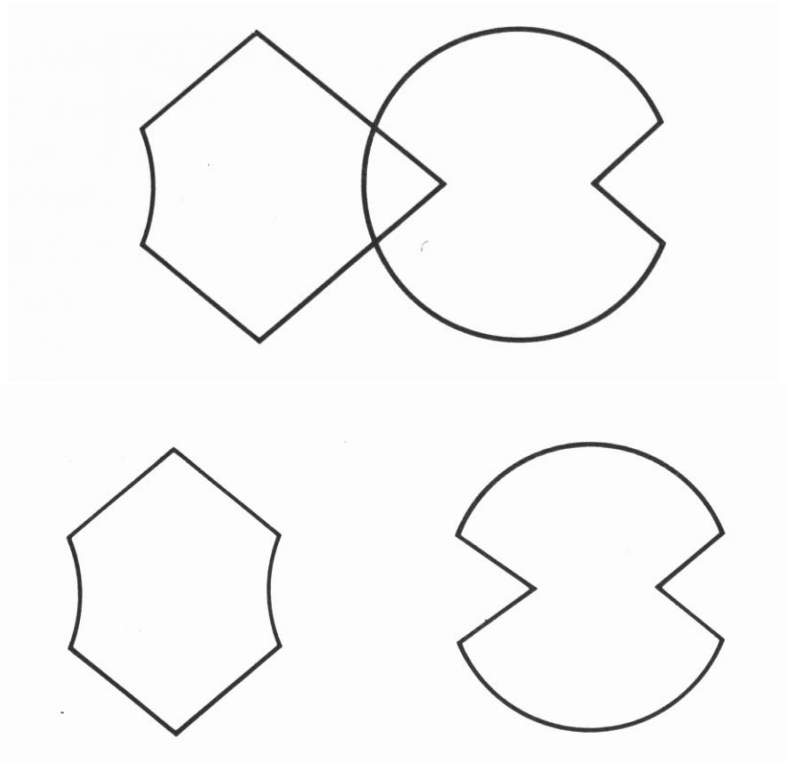
Gestalt Laws of Perceptual Organization

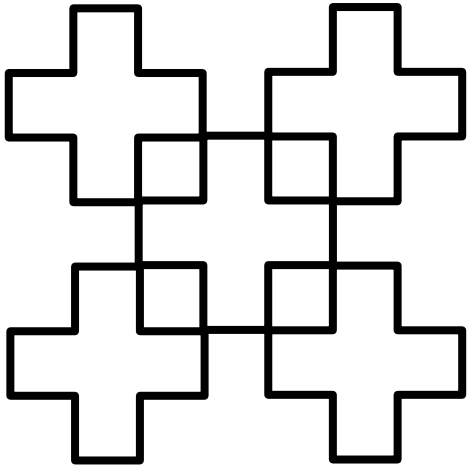


Symmetry

Perceptual Organization

Gestalt Laws of Perceptual Organization

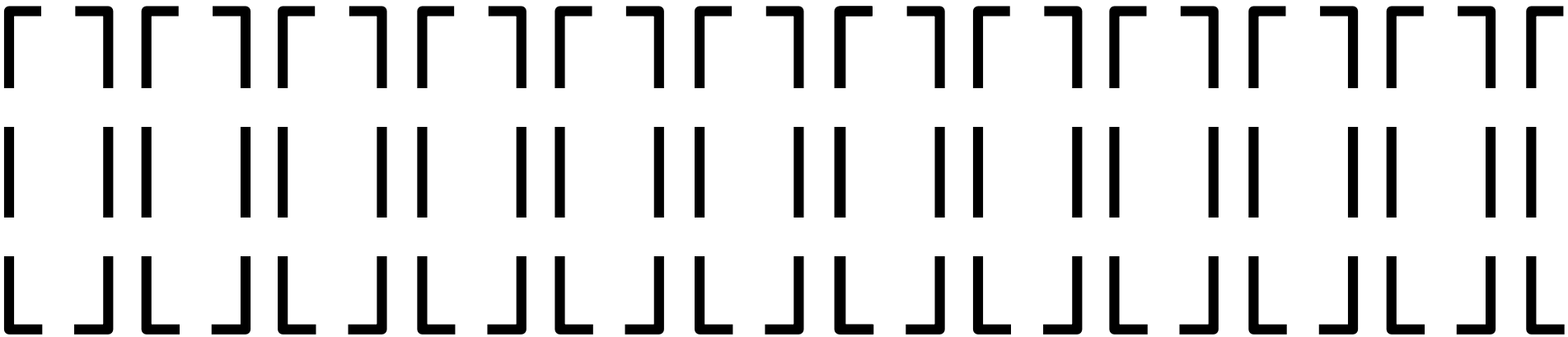




Factor of Direction (Good continuation)

Perceptual Organization

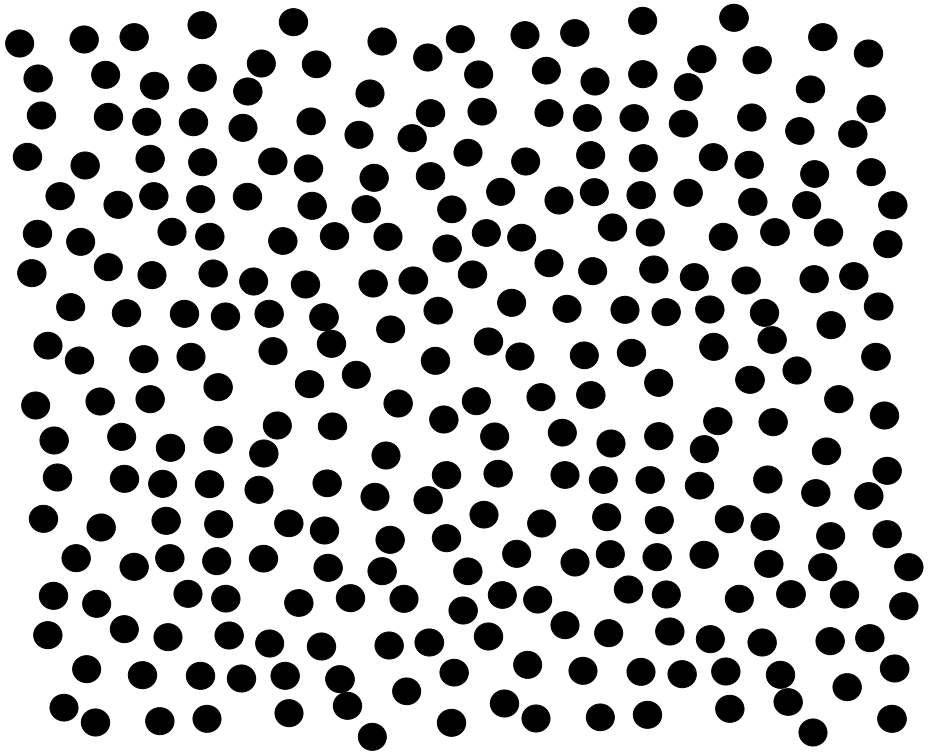
Gestalt Laws of Perceptual Organization



Closure

Perceptual Organization

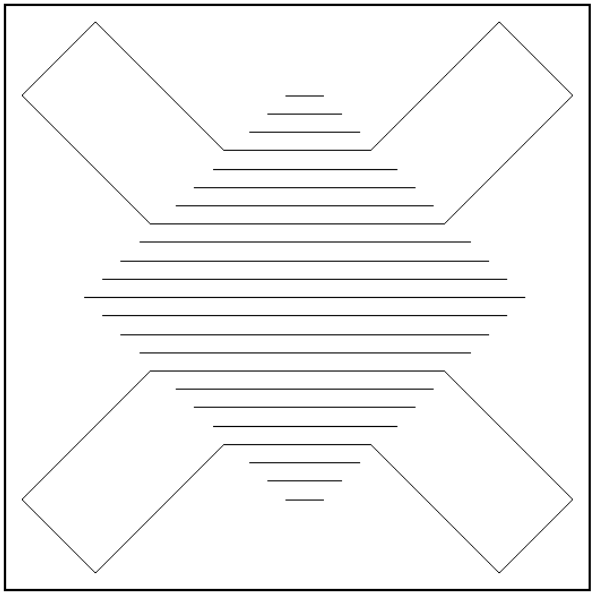
Gestalt Laws of Perceptual Organization

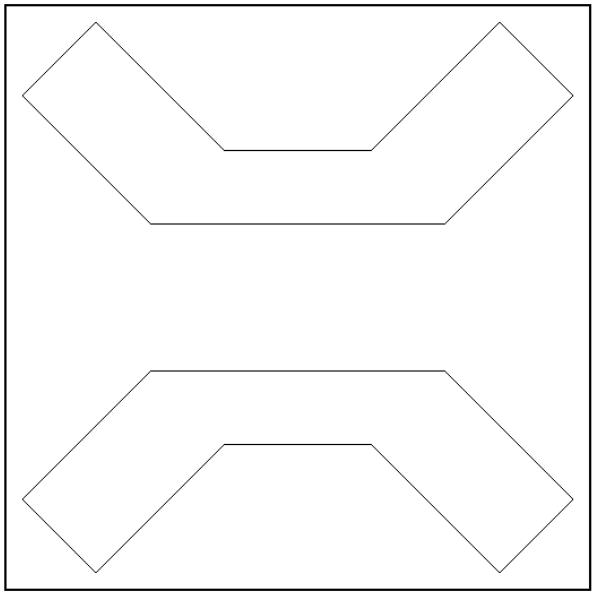


Common Fate

Perceptual Organization

Gestalt Laws of Perceptual Organization





Social conformity of a line

Grouping

Gestalt Laws of Perceptual Organization

And…Past experience

Perceptual Organization

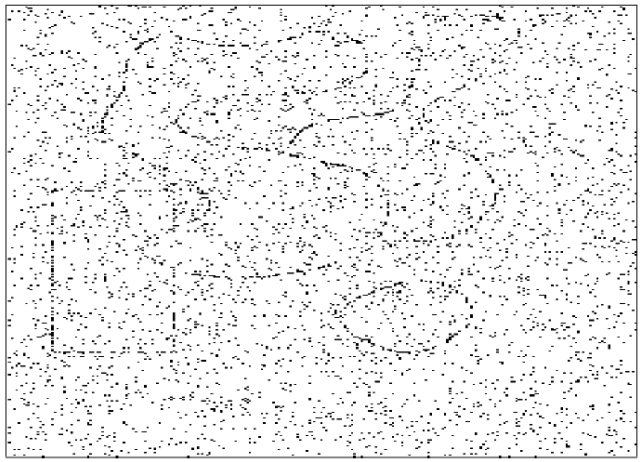
Gestalt Laws of Perceptual Organization

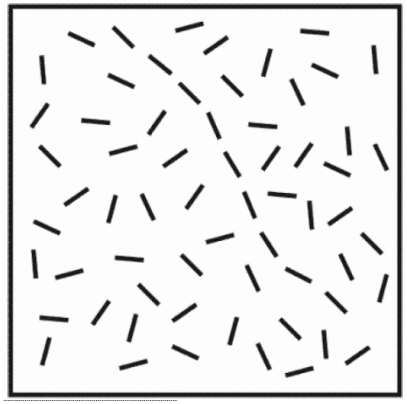
And the list goes on and on…

Perceptual Organization

Issues in Perceptual Organization

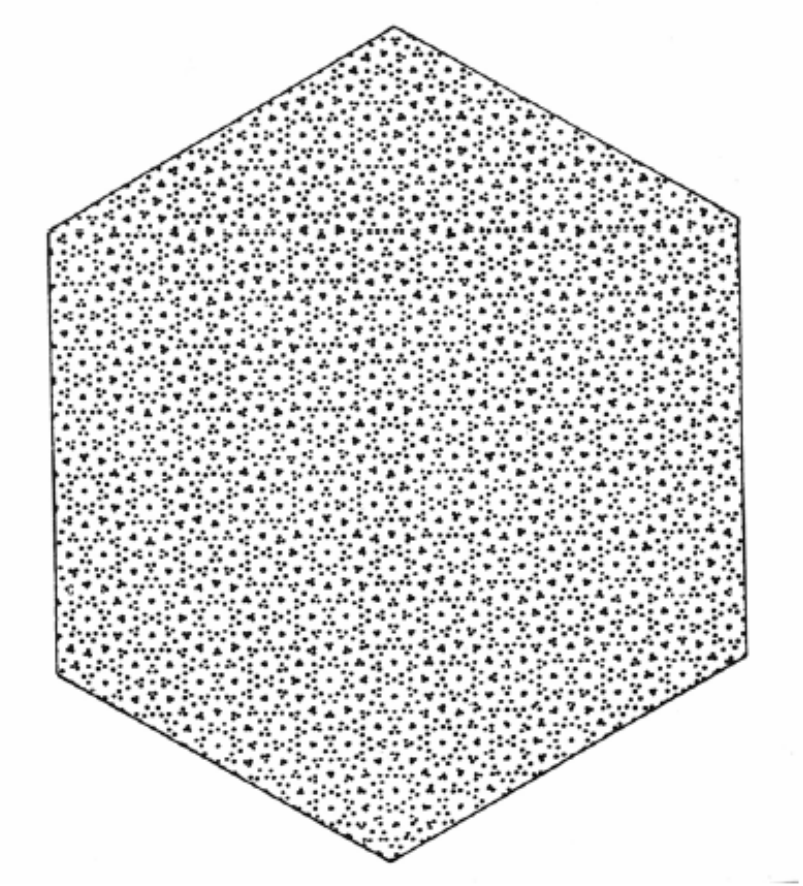
- **Grouping**
- **Segmentation**
- **Modal and Amodal Shape Completion**
- **Figure Ground Segregation**
- **Emergent Properties**
- **Multi-stable interpretations**





Perceptual Organization

Issues in Perceptual Organization



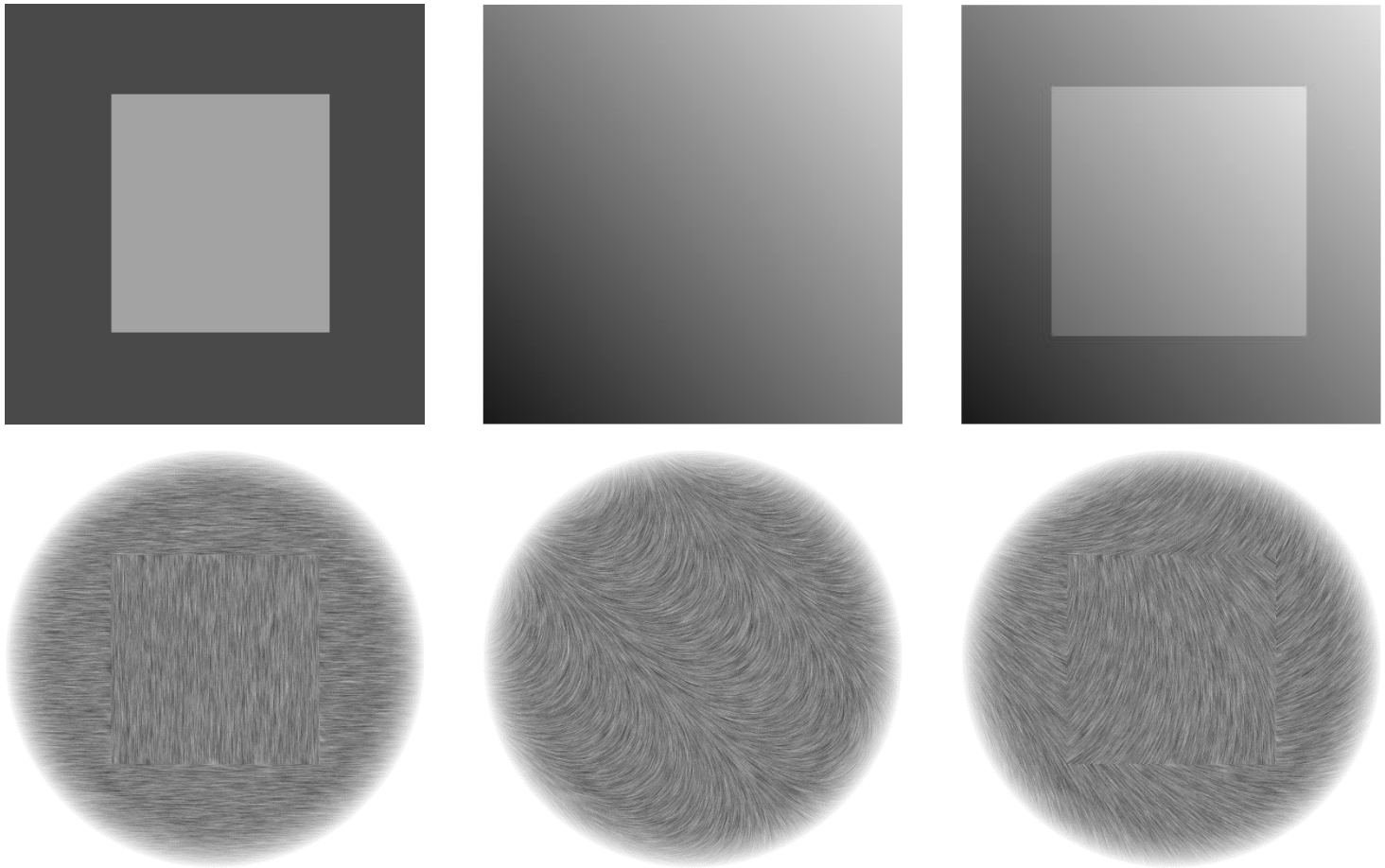
Perceptual Organization

Issues in Perceptual Organization

- **Grouping**
- **Segmentation**
- **Modal and Amodal Shape Completion**
- **Figure Ground Segregation**
- **Emergent Properties**
- **Multi-stable interpretations**

Perceptual Organization

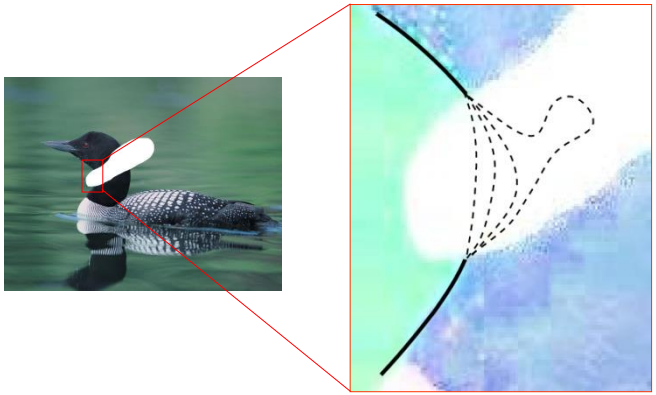
Issues in Perceptual Organization

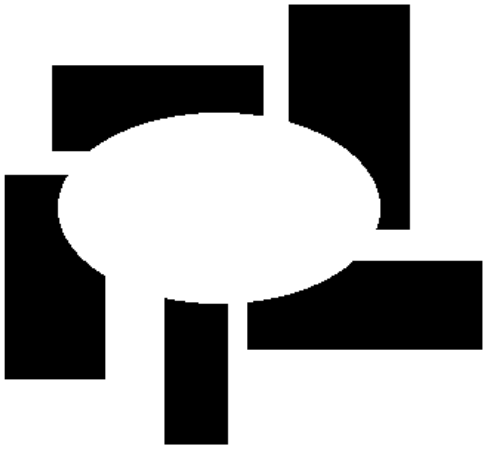


Perceptual Organization

Issues in Perceptual Organization

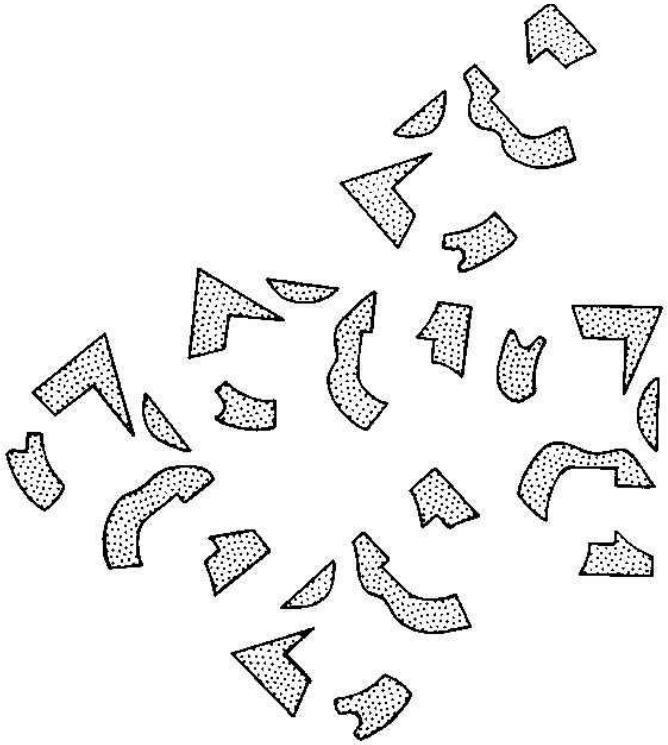
- **Grouping**
- **Segmentation**
- **Modal and Amodal Shape Completion**
- **Figure Ground Segregation**
- **Emergent Properties**
- **Multi-stable interpretations**





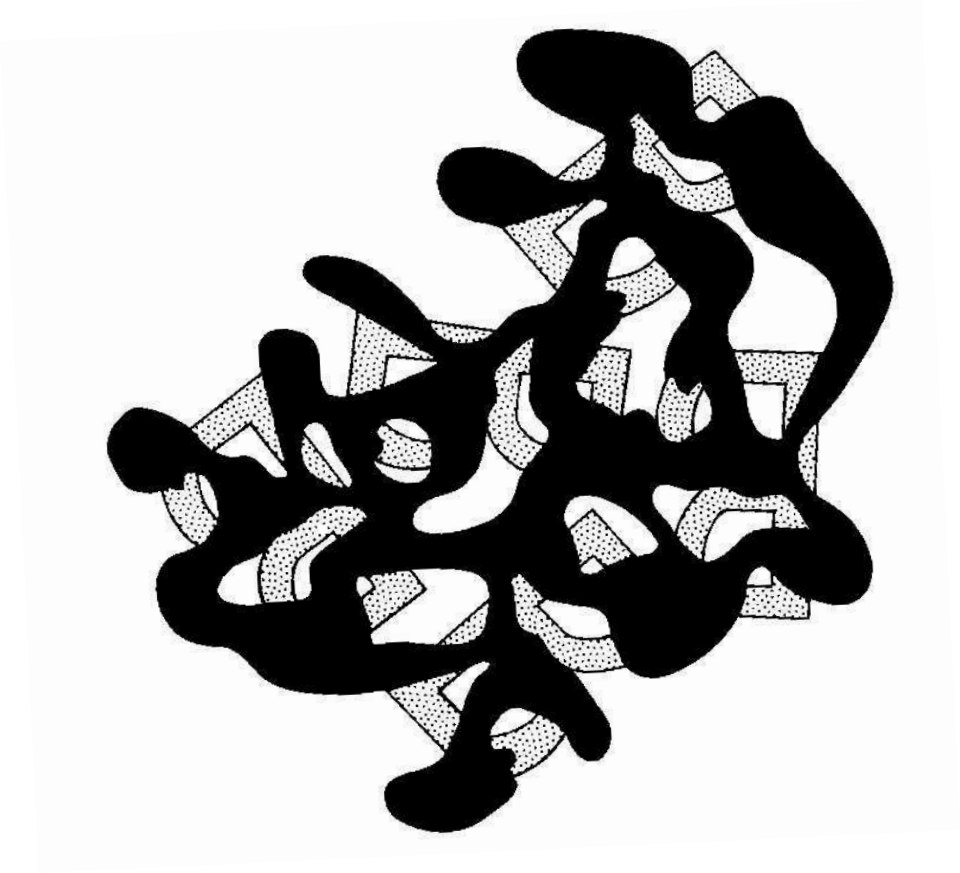
Perceptual Organization

Issues in Perceptual Organization



Perceptual Organization

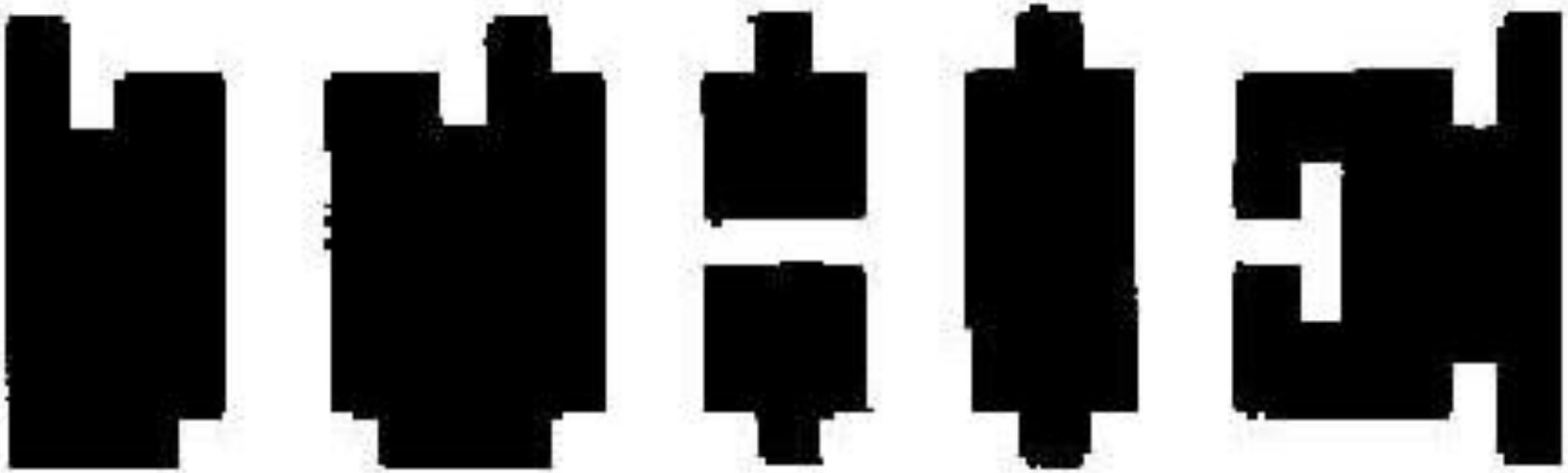
Issues in Perceptual Organization



Perceptual Organization

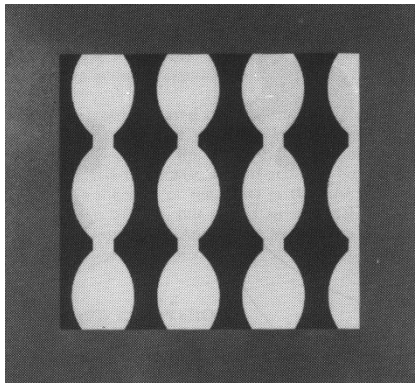
Issues in Perceptual Organization

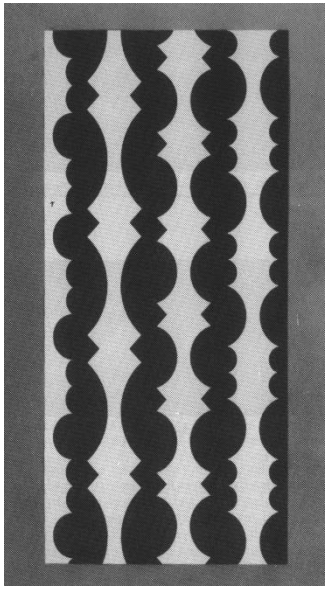
- **Grouping**
- **Segmentation**
- **Modal and Amodal Shape Completion**
- **Figure Ground Segregation**
- **Emergent Properties**
- **Multi-stable interpretations**

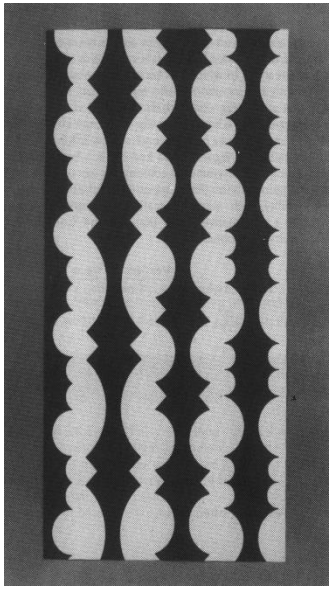


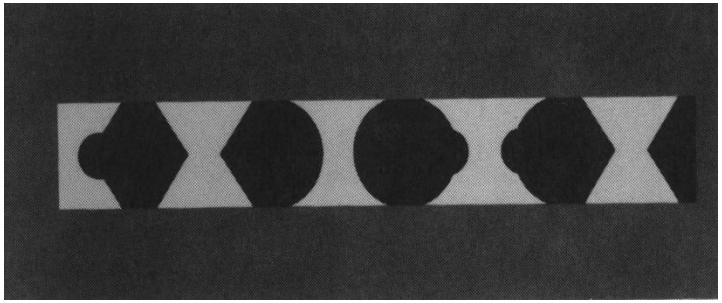
Perceptual Organization

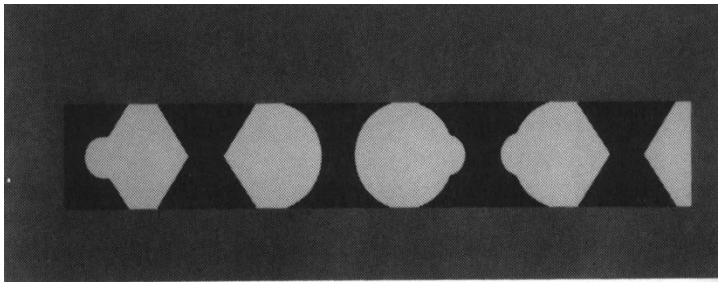
Issues in Perceptual Organization







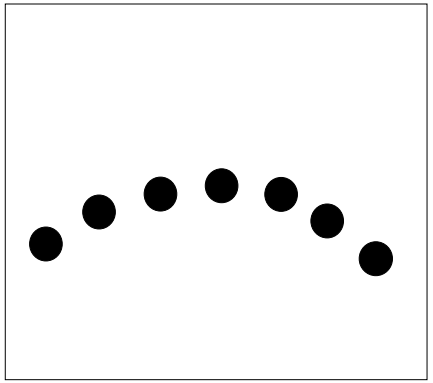


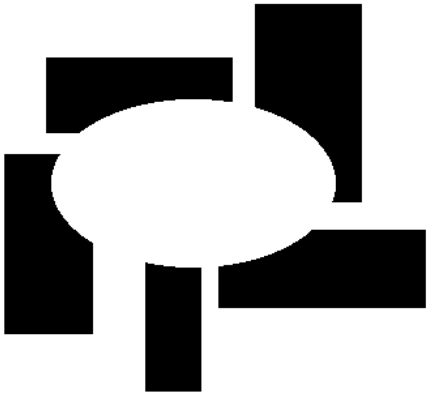


Perceptual Organization

Issues in Perceptual Organization

- **Grouping**
- **Segmentation**
- **Modal and Amodal Shape Completion**
- **Figure Ground Segregation**
- **Emergent Properties**
- **Multi-stable interpretations**

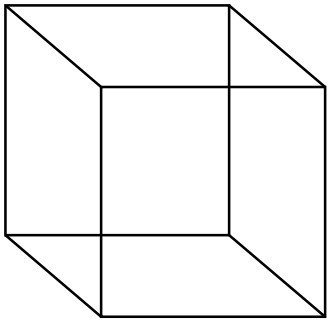




Perceptual Organization

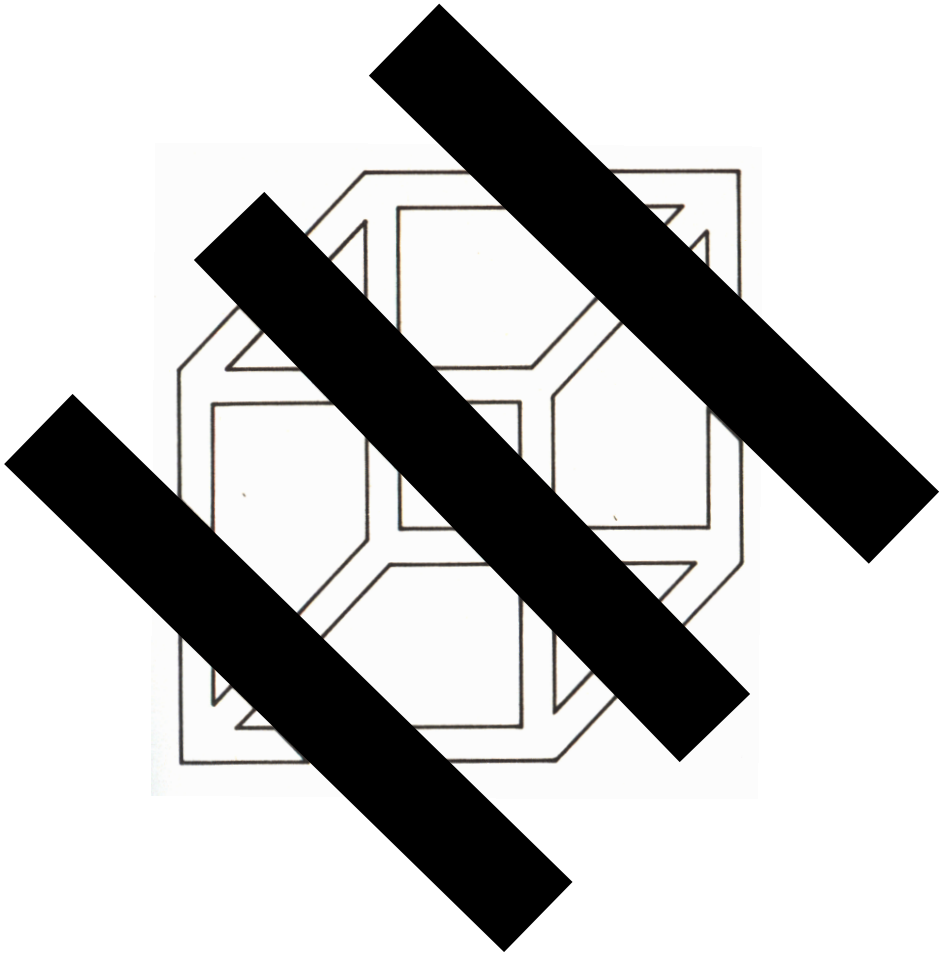
Issues in Perceptual Organization

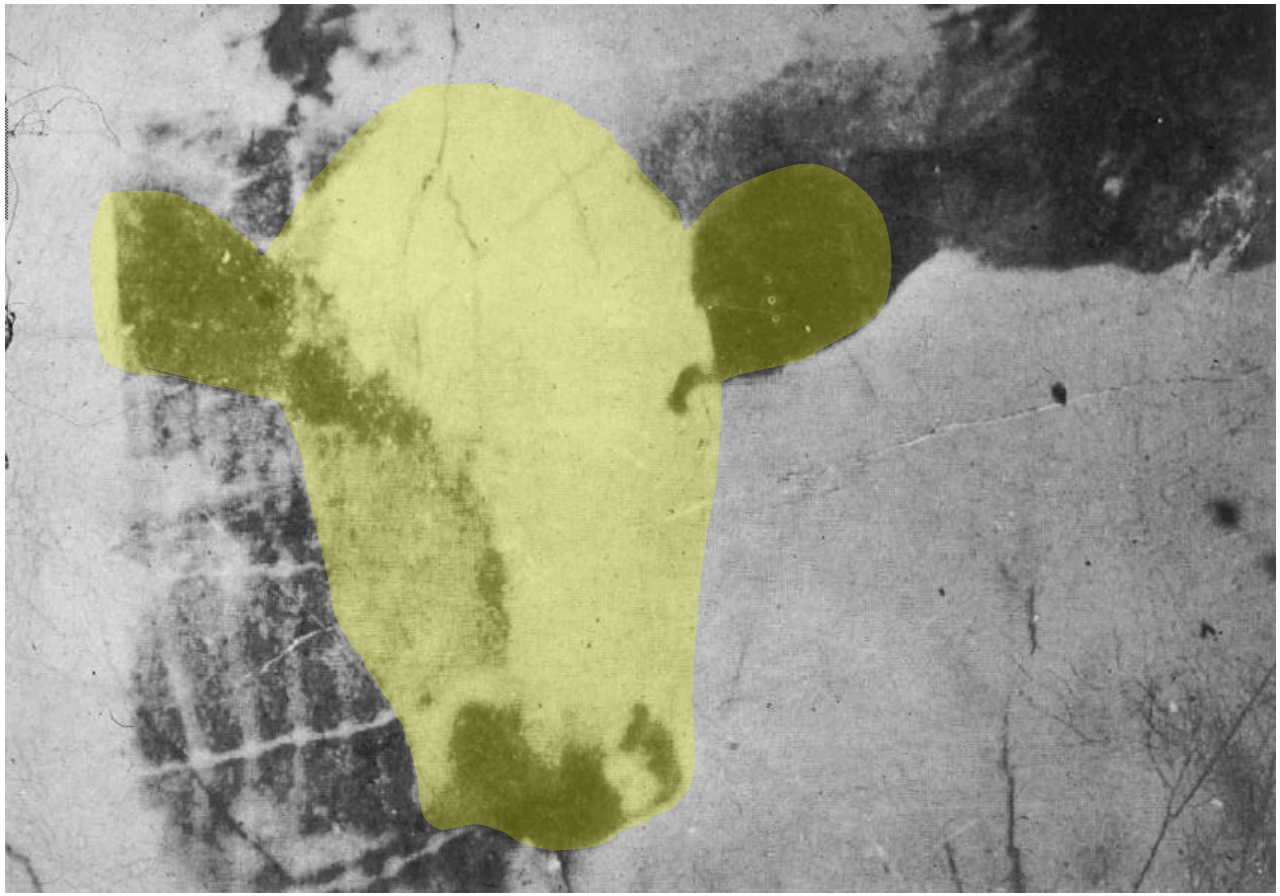
- **Grouping**
- **Segmentation**
- **Modal and Amodal Shape Completion**
- **Figure Ground Segregation**
- **Emergent Properties**
- **Multi-stable interpretations**



Perceptual Organization

So what's in here?



Perceptual Organization

…and here?